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Paleomagnetism of the Late Cretaceous ignimbrite from the Okhotsk-Chukotka Volcanic Belt, Kolyma-Omolon Composite Terrane: Tectonic implications

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ABSTRACT

New Late Cretaceous paleomagnetic results from the Okhotsk-Chukotka Volcanic Belt in the Kolyma-Omolon Composite Terrane yield stable and consistent remanent directions. The Late Cretaceous (86–81 Ma) ignimbrites from the Kholchan and Ola suites were sampled at 19 sites in the Magadan area (60.4° N, 151.0° E). We isolated the characteristic paleomagnetic directions from 16 sampled sites using an alternating field demagnetization procedure. The primary nature of these directions is ascertained by dual polarities and positive fold tests. A tilt-corrected mean direction ($D = 42.8°$, $I = 84.7°$, $k = 46.0$, $\alpha_{95} = 10.0°$) yields a paleomagnetic pole of 66.7° N, 168.5° E ($A_{95} = 18.8°$) which appears almost identical to the 90–67 Ma pole reported from the Lake El'gygytgyn area of the Okhotsk-Chukotka Volcanic Belt (Chukotka Terrane). This consistency suggests that the Kolyma-Omolon Composite Terrane and Chukotka Terrane has acted as a single tectonic unit since 80 Ma without any significant internal deformation. Accordingly, we calculate a combined 80 Ma characteristic paleomagnetic pole (Long. = 164.7° E, Lat. = 68.0°, $A_{95} = 10.9°$, $N = 12$) for the Kolyma-Omolon-Chukotka Block which falls 16.5–17.5° south of the same age poles from Europe and East Asia. We ascribe this discrepancy in pole positions to tectonic activity in the area and infer a southward displacement of 1640 ± 1380 km for the Kolyma-Omolon-Chukotka Block with respect to the North American and Eurasian blocks since 80 Ma; more than 260 km of it is attributed to tectonic displacement in the Arctic Ocean due to the opening of the Canadian Basin.
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1. Introduction

The present-day Eurasian Continent is a collage of several continental blocks formed as a consequences of sequential collisions and suturing. As shown in Fig. 1, the northeastern part of this mega continent comprises the Siberian Craton, Kolyma-Omolon Composite Terrane and Chukotka Terrane (Parfenov et al., 2009; Stone et al., 2009). Amalgamation of the Kolyma-Omolon Composite Terrane with Siberia in the Mesozoic has been described as one of the paramount tectonic events in the northeastern part of Eurasia (Parfenov et al., 2009). A boundary between the Kolyma-Omolon Composite Terrane and the Siberian Craton is characterized by prominent topographic features including the 1000 km long "S" shaped Verkhoyansk mountain range and the 1500 km long convex Chersky mountain range. These topographic expressions are analogous to those developed along the India-Asia and NCB-SCB collision zones. Thorough understanding at the scale of the Kolyma-Omolon Composite Terrane is required to elucidate collision tectonics in the northeastern part of Eurasia.

The Okhotsk-Chukotka Volcanic Belt (OCVB) of northeastern Eurasia, which is considered to be one of the largest volcanic

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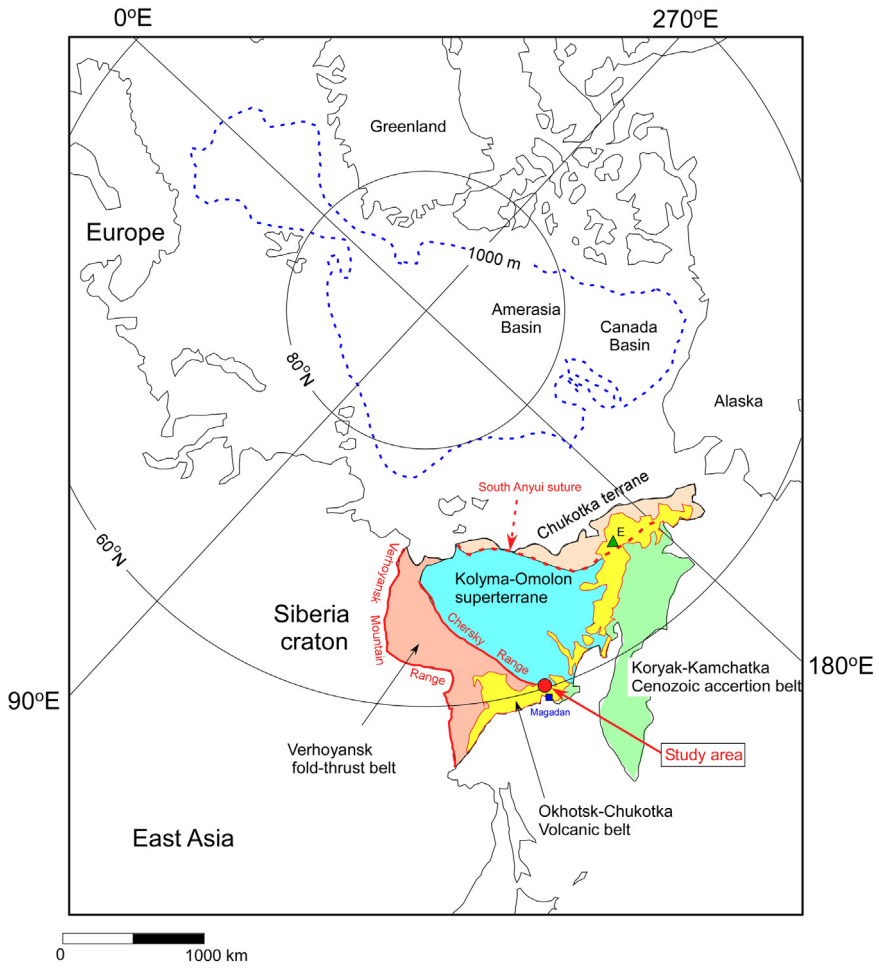


Fig. 1. (a) A simplified tectono-geographical map of the Arctic region. (b) Structural sketch map of the Kolyma-Omolon Composite Terrane and Chukotka Terrane (modified from Stone et al., 2009).

provinces in the world, extends from the Chukotka Terrane in the northeast via Kolyma-Omolon Composite Terrane to the Verhoyansk mountain range. According to some of the recent geological studies (Hourigan and Akinin, 2004; Belyi, 2008; Stone et al., 2009; Sakhno et al., 2010), this belt was formed as a consequences of late Cretaceous volcanic activities. Until a few years ago, some geologists believed that the OCVB is a subduction related belt (Hourigan and Akinin, 2004; Parfenov et al., 2009; Akinin and Miller, 2011). However, Tikhomirov et al. (2012) recently attributed it to silicic igneous province formed in the intra-plate environment. Although the origin of volcanism in this belt remains uncertain, their paleomagnetic signatures can help us to document the Late Mesozoic paleo-geographic framework of the Kolyma-Omolon Composite Terrane and the Chukotka Terrane.

Accordingly this paper presents new paleomagnetic data from the Kolyma-Omolon Composite Terrane because paleomagnetism can provides a quantitative record of block rotation and N-S displacement (Fig. 1). For this purpose we sampled volcanic rocks exposed in the Magadan area (60.4° N, 151.0° E) of the OCVB.

2. Geologic setting

As described by Parfenov et al. (2009) the northeastern part of Eurasia (east of the Siberian Craton) is made up of several terranes; the Verkhoyansk fold and thrust belt, Kolyma-Omolon Composite Terrane, Chukotka Terrane and Koryak-Kamchatka Cenozoic accretionary belt (Fig. 1). These terranes were amalgamated as a result of sequential collisions and accretions during the Mesozoic and Cenozoic. Formation of the Mesozoic Verkhoyansk fold and thrust belt, as a result of the collision of the Siberian Craton with the allochthonous Kolyma-Omolon Composite Terrane, was followed by the Chukotka Terrane accretion (Oxman, 2003). An extensive intra-plate magmatic activity from the Early to Late Cretaceous (from 106 ± 1.7 to 74 ± 1.2 Ma), which brought widespread volcanic emplacement over the Mesozoic assemblages of the Kolyma-Omolon and Chukotka terranes, has been considered to be the cause of OCVB formation (Tikhomirov et al., 2012). The Cenozoic Koryak-Kamchatka accretionary belt formed as a final phase of this sequence.

The Arman volcanic field in the Magadan area forms the central part of the Okhotsk sector in the OCVB (Hourigan and Akinin, 2004). These volcanics were emplaced over deformed sedimentary rocks of the Momoltykich and Khasyn suites deposited in a passive margin back-arc basin (marine environment) from the Permian to the Jurassic. As shown in Fig. 2, the Arman volcanic field is divided into six suites; the Arman, Narauli, Kholchan, Ulyn, Ola and Mygdykit in ascending order (Anorov et al., 2013a,b). Based on floral assemblages (the Arman flora) found in the type area late Albian and Cenomanian ages have been assigned to the Arman volcanic field. The $^{40}Ar/^{39}Ar$ dating assigns an age of 85.5–85.6 Ma to the Narauli Suite, 81.7–83.6 Ma to the Ola Suite and 74.0–78.8 Ma to the Mygdykit Suite. Hence, Late Cretaceous age is assigned to the Arman volcanic field.

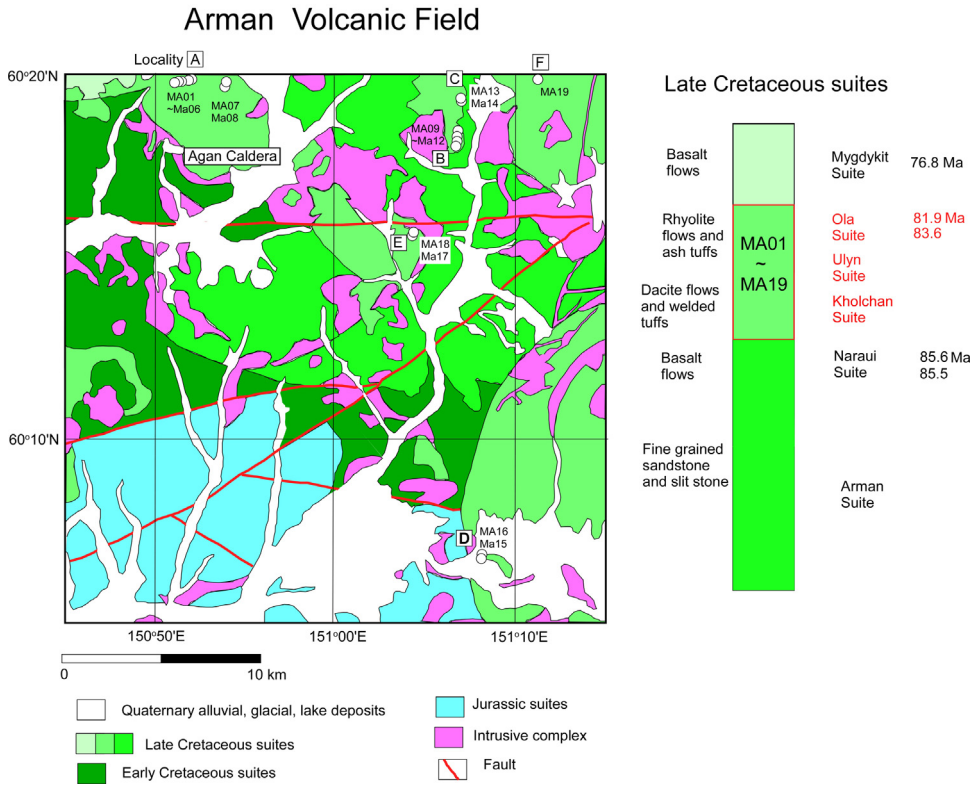


Fig. 2. Geological map of the study area (Magadan), a part of the Arman volcanic field in the OCVB (modified from Anorov et al., 2013a,b). Lithostratigraphic columns are also shown together with $^{40}Ar/^{39}$ Ar ages (Hourigan and Akinin, 2004). Sampling sites are shown by closed circles. A–F: sampling localities.

The volcanic rocks of the Kholchan and Ola suites which yields $^{40}Ar/^{39}Ar$ ages of 81–86 Ma were sampled at 19 sites for this paleomagnetic study (Fig. 2). The Kholchan suite consists of rhyolite-dacite tuffs, welded tuffs and volcanic flows with local lenses of tuffaceous sandstones, conglomerate and siltstones. Conversely, the Ola suite consists of non-welded to moderately welded ash-flow tuffs and tabular flows of crystal rich rhyolite. The sampled sites are located near the Agan Caldera, on the north side of the city of Magadan (60.4° N, 151.0° E). Approximately 8–11 paleomagnetic block samples, from 3 to 14 m in covered area, were collected at each site. Orientation of each sample was made by magnetic compass, because a lack of sunshine in the study area prevented us from using a sun-compass. Bedding attitude at each sampling site was measured using the eutaxitic structure of the volcanic rocks (stretched pumice lineation and aligned phenocrysts). Because an eutaxitic structure cannot always indicate a paleohorizontal plane, samples were collected from widely spaced outcrops to reduce the expected error from uncertain tectonic correction. The present day geomagnetic field declination at each sampling site was noted from International Geomagnetic Reference Field (IGRF) (the International Association of Geomagnetism and Aeronomy Working Group V-MOD, 2010).

3. Laboratory procedures

Cylindrical specimens of 25 mm in diameter and 22 mm in length were prepared from each block sample in the laboratory. Natural remanent magnetizations (NRMs) were measured using the Kobe University based spinner magnetometer (Natsuhara Giken SMM-85). One specimen per sample was subjected to stepwise thermal demagnetization (ThD) up to 620 °C using a "Natsuhara Giken TDE-91C" thermal demagnetizer with a residual field of less than 5nT. To obtain reliable paleomagnetic results, duplicate specimens were demagnetized by alternating field demagnetization (AFD) using a "Natsuhara Giken DEM93" AF demagnetizer. Demagnetization results were plotted on orthogonal vector diagrams (Zijderveld, 1967) to assess component structure and on equal-area projections to evaluate directional stability during demagnetizations. The remanent directions were identified using principal component analysis (Kirschvink, 1980) and the great circle method (Halls, 1978; McFadden and McElhinny, 1988). Site-mean directions were calculated using Fisherian (Fisher, 1953) and McFadden and McElhinny statistics (McFadden and McElhinny, 1988).

4. Results

4.1. Rock magnetism

Samples from five ignimbrite layers corresponding to sites MA01, MA02, MA05, MA10 and MA16 were analyzed for rock magnetic characteristics. Progressive acquisition of isothermal remanent magnetization (IRM) was performed on representative specimens up to a maximum field of 2.0 T using a Magnetic Measurements Pulse Magnetizer (MMPM-10). Thermal demagnetization of the composite IRMs (imparted at 2.0 T, 0.4 T and 0.12 T along the *z*, *y* and *x* axes of the specimens, respectively) was conducted to detect the unblocking temperature spectra of the analyzed specimens (Lowrie, 1990). The IRM acquisition curves (Fig. 3) show complete saturation up to a 0.3 T applied field yielding magnetite as a dominant remanence carrier. Thermal demagnetization of the composite IRMs (Fig. 3) indicates an unblocking temperature spectra of the soft component at 580 °C, confirming the presence of magnetite.

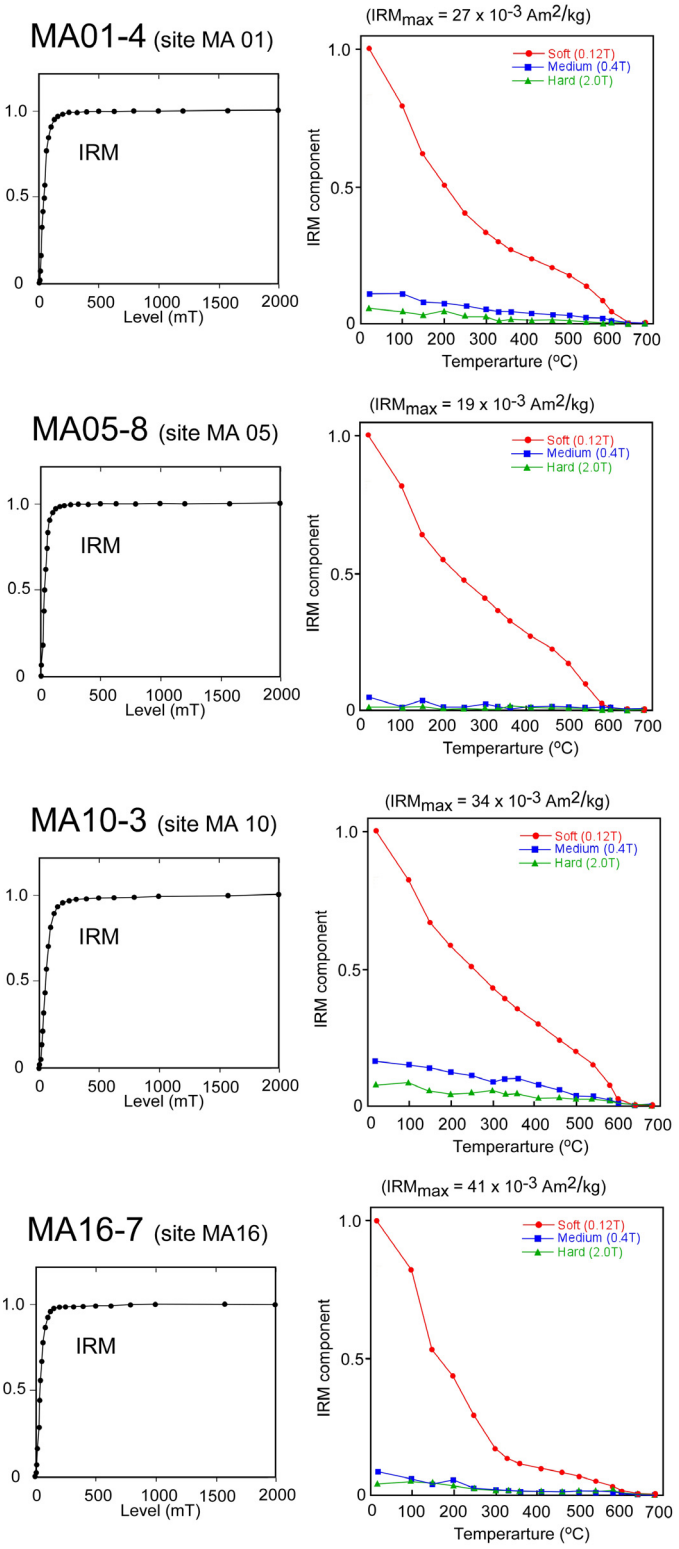


Fig. 3. Rock magnetic results of selected specimens from Kholchan and Ola volcanic suites. The IRM acquisition and thermal demagnetization of three-axes IRMs imparted by DC fields of 2.0 T, 0.4 T and 0.12 T along three perpendicular axes.

4.2. Paleomagnetism

4.2.1. The NRM intensity

A wide range of initial NRM intensities between 0.05 A/m and 27.4 A/m is observed in the analyzed samples although all of them are collected from ignimbrite (Fig. 4). Intensities higher than 8 A/m are observed in 42 out of 250 samples from sites MA03, MA05, MA09, MA10, MA12, MA13, MA14 and MA18. However, mean intensities lower than 5.8 A/m are observed in five of these sites (MA03, MA10, MA12, MA14 and MA18) and exclusively high intensities are observed in most samples of the other three sites (MA05, MA09 and MA13). The remaining 72% of the analyzed samples have intensities ranging between 0.17 A/m and 3.6 A/m.

4.2.2. Demagnetization behavior during ThD and AFD analyses

Demagnetization behavior of the analyzed samples during progressive ThD depends very much on their NRM intensities. In general, samples with NRM intensities larger than 5.0 A/m reveal a single component behavior and a gradual decrease toward the origin without any visible change in direction (Fig. 5). An unblocking temperature of approximately 590 °C is observed in such specimens (Fig. 5). Coversely, two components behavior is generally observed in samples yielding less than 1.7 A/m NRM intensity. Low and high temperature component magnetizations in the two components are unblocked at approximately 250 °C and 590 °C, respectively. Because the characteristic remanent magnetization (ChRM) is unblocked by 590 °C, magnetite is a main magnetic carrier. Two components behavior observed by ThD analyses is also confirmed by AFD analyses. However, in some samples the AFD analysis identifies two NRM components where only one component is observed by ThD analysis (e.g., site MA05 in Fig. 5 and MA01 in Fig. 6).

Generally, identical NRM directions are observed in most samples when analyzed either by the ThD or AFD procedures (Fig. 5; Table 1), however, several samples from site MA01 yield different directions treated by these procedures (Fig. 6). Two samples from site MA01 (MA01-1 and MA01-4) with initial NRM intensities over 1.7 A/m reveal a southwestward declination and shallow inclination prior to demagnetization. Very little change in their directions appears when analyzed by the ThD procedure. However, significant change in direction to northerly declination and steep positive inclination appears when analyzed by the AFD procedure.

4.2.3. The NRM directions

Because the AFD procedure proved more effective than the ThD in randomizing the secondary remanent magnetizations, results from the former procedure are used here to detect the ChRM directions. The NRM directions are mostly identified by principal component analysis, however, the great circle method is also used to determine directional behavior in 11 out of 123 samples. Among them, two samples are from site MA01 (Fig. 6), seven from site MA02 and two from site MA09. Because of very strong NRM intensities (24.9 and 8.6 A/m) two samples of site MA14 are excluded from final site mean calculations. The results of principal component analysis and great circle method are listed in Table 1.

The site-mean directions from 16 out of 19 sampling sites reveal precision parameters (k) larger than 10. Relatively small values of k (<6.1) appears in the remaining three sites (MA03, MA05 and MA13) with large NRM intensities (>50 × 10^{-1} A/m) which forced us to exclude them from the final calculations. Typically larger k values are observed at sites with lower NRM intensities (Fig. 4b). However, large k values appeared at two high intensity sites (MA09 and MA14), which could be due to the use of the great circle method and the exclusion of some very high intensity samples. Accordingly, 16 site-mean directions with k values over 10 are illustrated on an equal-area projection (Fig. 7a). Following tilt corrections 14 out of 16 sites yield normal polarity directions with steep downward inclinations (between 54.5° and 86.7°) and the remaining two sites (MA15 and MA16) reveal reversed polarity directions with steep upward inclinations (∼−66.2°). The Late Cretaceous ignimbrites of

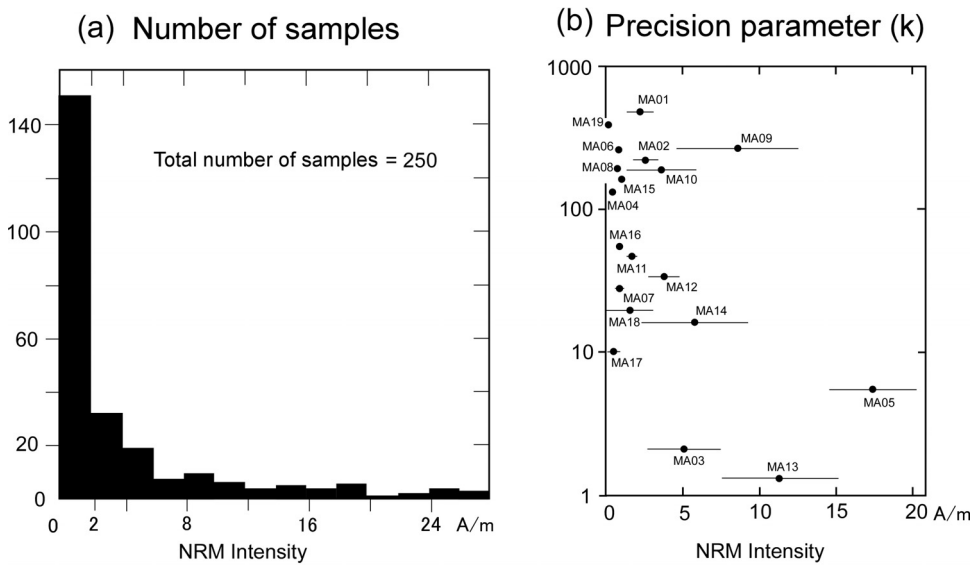


Fig. 4. (a) The NRM intensity distribution in the analyzed samples from Okhotsk-Chukotka Volcanic Belt. As indicated, the intensities before demagnetization are in a range between 0.05 A/m and 27.4 A/m. (b) Precision parameter (k) versus mean NRM intensity for each site. Small dispersions appear in the sampling sites with weaker NRM intensity.

the Kholchan and Ola suites thus keep a record of both normal and reversed polarity directions.

4.2.4. Formation mean direction

As a first option, we calculated formation mean direction on the basis of 16 site mean directions (Fig. 7a) which yield a mean direction of $D = 43.0^\circ$, $I = 76.0^\circ$ ($\alpha_{95} = 8.1$) in geographic coordinates. Assuming the attitude of eutaxitic texture as a paleohorizontal plane a tilt-corrected formation mean direction of $D = 17.6^\circ$, $I = 82.5^\circ$ ($\alpha_{95} = 10.6$) is calculated. Although both of these directions are indistinguishable from each other at the 95% confidence limit (Table 1), relatively large dispersion appears in a tilt-corrected mean direction ($k = 13.1$) compared to an in situ mean direction ($k = 21.8$). The McElhinny (1964) fold test applied to these data sets proved inconclusive. However, the correlation test of McFadden (1990) and the DC tilt test of Enkin (2003) produced negative results. Bearing in mind a record of both normal and reverse polarities in the studied rocks the negative fold tests may not be due to remagnetization but rather to inaccurate measurement of the paleohorizontal plane (Lyons et al., 2002). The outcrop-scale eutaxitic structure, which has been used as a paleohorizontal plane in several studies, generally represent an initial dip of flow emplacement over irregular topography, paleohill slopes or paleostream channels (McIntosh, 1991; Hagstrum and Gans, 1989; Chapin and Lowell, 1979). However, such outcrop scale structures in ignimbrite cannot always be used as a true substitute for a large scale paleohorizontal plane.

Bearing in mind this uncertainty we calculated mean directions on locality basis as an alternative option (A to F in Fig. 1 and Table 2) where a unit weight is added to each locality mean before computing an in situ formation mean direction ($D_g = 65.6^\circ$, $I_g = 75.0^\circ$, $k = 23.3$, $\alpha_{95} = 14.2^\circ$, $N = 6$) and a tilt corrected formation mean direction ($D_s = 42.8^\circ$, $I_s = 84.7^\circ$, $k = 46.0$, $\alpha_{95} = 10.0^\circ$, $N = 6$) (Fig. 7b). These formation mean directions are shown in Fig. 7b. We adopted this procedure to randomize an expected error caused by outcrop scale variations in paleohorizontal features within each locality. This procedure yields an increase in the k parameter from 23.3 to 46.0 after tilt correction and confirmation of a positive fold test at the 95% confidence limit (McElhinny, 1964). Application of the DC tilt test (Elkin 2003) to our locality mean direction also produces a positive test at $78 \pm 63\%$ unfolding. As an additional check, we also applied the second definition (ξ_2) of the McFadden method (McFadden, 1990; Cogne, 2003) which confirms a positive fold test at the 95% confidence level. This test reveals a critical value of 2.862 for parameter ξ_c and 3.362 (1.635) for ξ_2 in geographic coordinates (stratigraphic coordinates). An apparent positive fold test results from the second option (localities mean direction), and thus suggests a pre-folding origin for the high-unblocking temperature component in the studied ignimbrites and is accordingly interpreted as the Late Cretaceous ChRM direction in the Magadan area. Because the Late Cretaceous (86–81 Ma) paleomagnetic poles calculated here both on the basis of locality-mean (66.7° N, 168.5° E, $A_{95} = 18.8^\circ$) and site mean (72.4° N, 165.7° E, $A_{95} = 18.9^\circ$) appear almost identical (Table 3) the earlier one is preferentially chosen as representative of the Magadan area.

5. Discussion

5.1. Reliability of the NRM directions

As noted above, we used a magnetic compass to orient paleomagnetic samples in the field and IGRF declination as a reference for each sampling site. To check for deflection of magnetic compasses near high intensity rocks, Gilder et al. (2003) have paleomagnetically studied basaltic rocks of western Tarim Basin (75°20′ E, 40°17′ N). For this study rock samples with NRM intensity up to 8 A/m were oriented both with magnetic and sun compasses. Compared to the IGRF declination of 3.9° in the study area, an in situ declination of $3.4 \pm 2.3^\circ$ was obtained from both types of orientations This comparison revealed negligible amounts of deflection in declination due to high (up to 8 A/m) intensity volcanic rocks and confirmed the reliability of IGRF declination in the study area. Because 72% of paleomagnetic samples for this study are collected from ignimbrites with less than 3.6 A/m NRM intensity, their orientations by magnetic compass are considered reliable. At high latitude the IGRF declination may deflect significantly with respect to local declination. However, a personal communication from the Magadan Geophysical Observatory (60.051° N, 150.728° E, 226 m) indicates an average declination of −13.37° for 2013, and the local IGRF shows an average declination of −11.0° for the same period (i.e., no significant error between two sets of orientations).

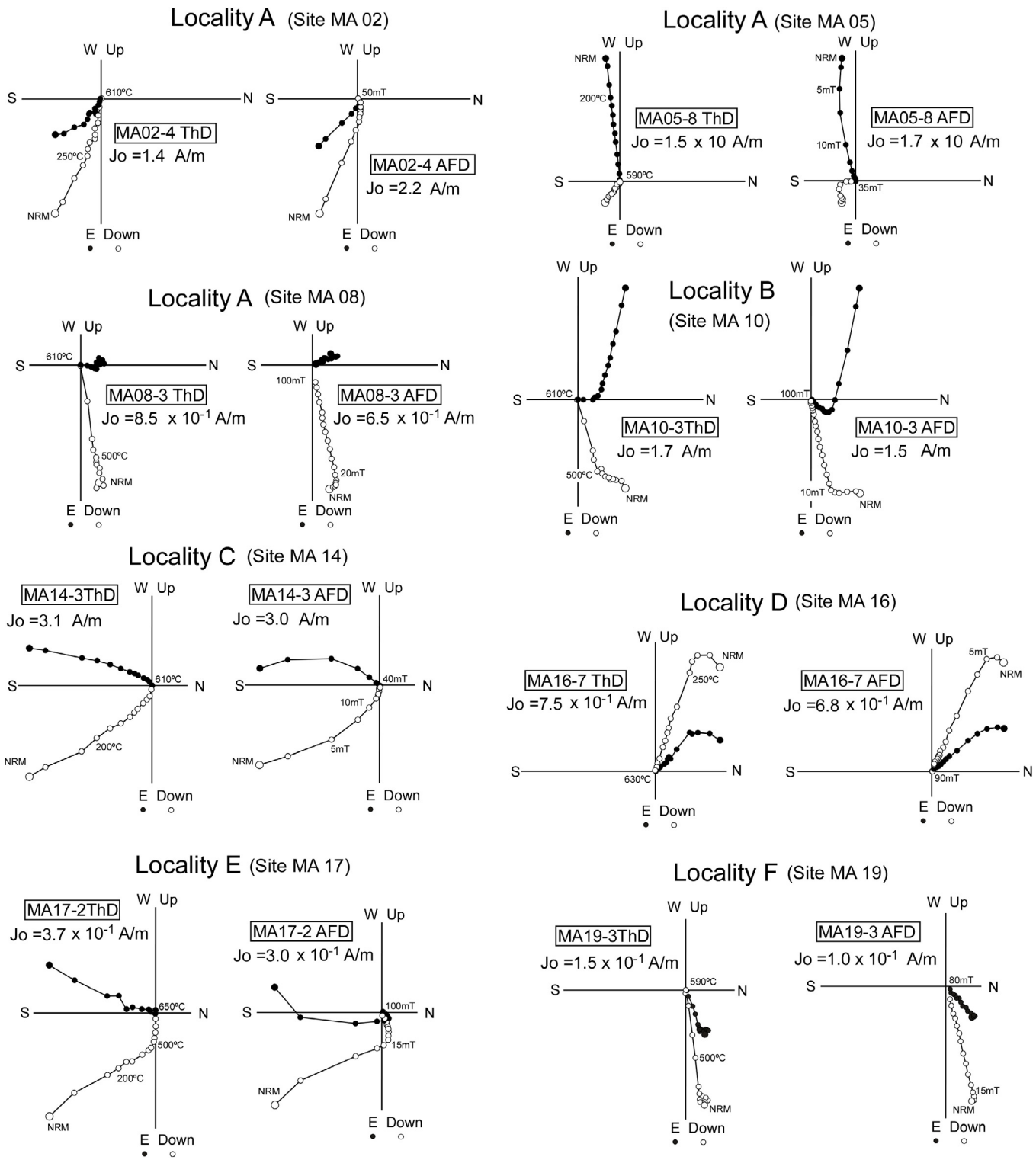


Fig. 5. In situ vector end-point diagrams for the representative ignimbrite specimens from six localities (A–F) after thermal demagnetization. Solid (open) symbols are projections onto the horizontal (vertical) plane.

Magnetic compass and IGRF provide an accuracy of orientation at ~5° in the Magadan area.

As noted in the results section, samples from three sites (MA03, MA05 and MA13) which carry NRM intensities higher than 5 A/m reveal single component behavior during ThD analyses. The site-mean NRM directions from these sites yield large dispersion (k < 6.1) compared to the remaining 16 sites. An exceptionally high NRM intensity and excellent directional stability in these sites suggest their complete remagnetization by lightning strike (Dunlop and Özdemir, 2001), while all other low intensity sites appear to have been partially remagnetized. As mentioned by Dunlop and Özdemir (2001), rocks partially remagnetized by lightning strike could be more successfully demagnetized by AFD procedure compared to ThD. Accordingly, the AFD data from six localities are used

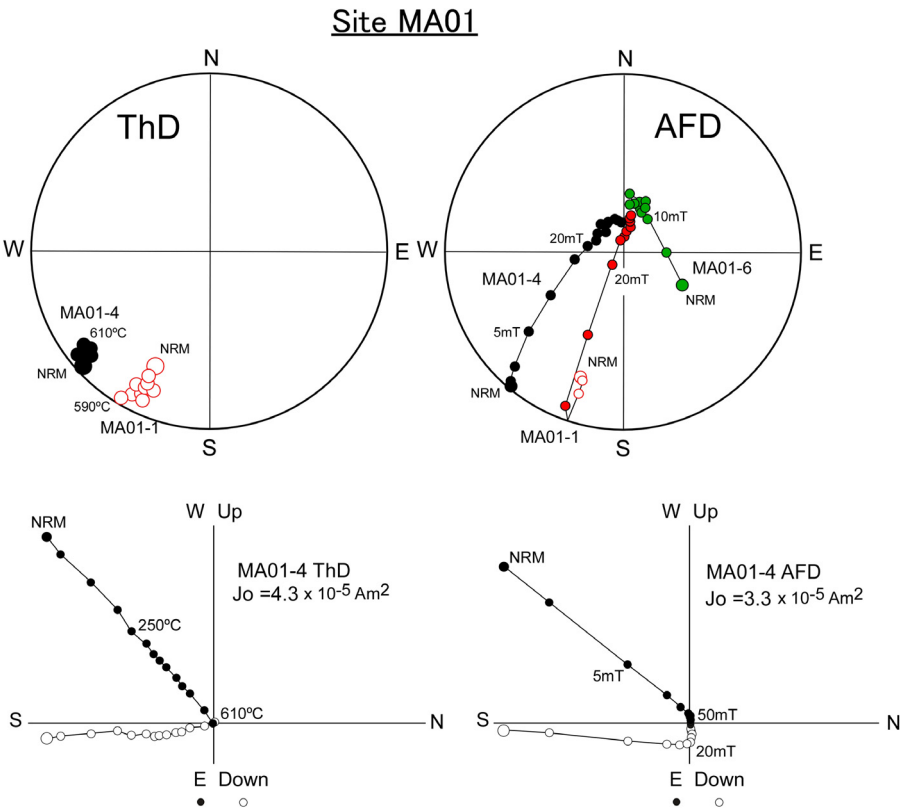


Fig. 6. Equal-area projections and vector end-point diagrams of representative specimens from site MA01; after ThD (left) and AFD (right). In equal-area projections, solid (open) symbols are projections onto the lower (upper) hemisphere. In the case of vector end-point diagrams, solid (open) symbols are projections onto the horizontal (vertical) plane.

Table 1
Paleomagnetic results of the ignimbrite samples collected from Late Cretaceous Kholchan and Ola suites (the Magadan area).

Sampling site	Latitude	Longitude	Strike (°)	Dip (°)	*n*	Intensity $\times 10^{-6}$ Am^2	Max	*n*	In situ		Tilt corrected		*k*	α_{95} (°)
									Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)		
MA01	60°19.80′	150°51.95′	209.0	10.0	11	22.4 ± 17.9	58.5	8	15.9	63.9	358.3	60.1	481.2	2.6
MA02	60°19.78′	150°52.03′	209.0	10.0	11	25.5 ± 15.8	50.6	8	16.3	65.6	357.4	61.8	225.7	4.2
MA03*	60°19.78′	150°51.85′	209.0	10.0	14	50.4 ± 46.7	151.8	6	101.1	3.5	100.4	12.3	2.1	62.4
MA03*								**7**	**104.9**	**−9.2**	**105.3**	**−0.2**	**2.1**	**56.4**
MA04	60°19.76′	150°51.85′	357.0	18.0	9	3.8 ± 1.7	6.5	8	19.1	67.4	46.7	56.4	134.1	4.8
MA05*	60°19.73′	150°51.64′	49.0	12.0	18	173.6 ± 56.5	278.0	9	299.8	28.8	295.4	38.9	5.5	24.2
MA05*								**9**	**308.6**	**31.0**	**305.2**	**42.1**	**6.1**	**22.6**
MA06	60°19.71′	150°51.65′	49.0	12.0	9	8.6 ± 2.6	13.5	8	37.7	70.7	71.3	69.5	259.4	3.4
MA07	60°19.72′	150°53.45′	107.0	27.0	18	9.0 ± 5.7	24.7	10	223.8	81.2	212.1	54.5	28.0	9.3
MA07								**6**	**234.1**	**81.6**	**214.5**	**55.2**	**33.2**	**11.8**
MA08	60°19.70′	150°53.42′	107.0	27.0	9	7.6 ± 1.7	9.9	8	349.1	78.1	215.6	72.3	195.4	4.0
MA09	60°18.18′	151°06.71′	189.1	3.0	9	85.7 ± 78.5	243.4	6	15.0	63.3	9.0	63.5	269.0	4.3
MA10	60°18.31′	151°06.71′	214.1	12.0	16	36.4 ± 45.4	187.0	8	24.2	71.1	359.3	64.4	192.0	4.0
MA10								**7**	**3.4**	**72.2**	**345.0**	**62.8**	**11.0**	**19.0**
MA11	60°18.15′	151°06.78′	187.1	22.0	9	17.3 ± 7.2	33.3	8	5.3	72.8	315.9	61.9	47.3	8.1
MA12	60°18.00′	151°06.67′	99.1	20.0	16	33.0 ± 20.0	88.4	7	336.5	78.5	234.5	75.6	34.5	10.4
MA12								**8**	**348.9**	**73.0**	**258.6**	**79.8**	**113.8**	**5.2**
MA13*	60°19.29′	151°06.56′	277.1	8.0	20	112.5 ± 74.8	274.6	11	69.6	23.4	67.3	18.3	1.3	74.1
MA14	60°19.26′	151°06.55′	169.1	11.0	12	57.5 ± 68.8	249.3	5	105.1	72.4	136.6	81.0	16.4	19.5
MA15	60°07.07′	151°08.00′	264.1	32.0	9	9.9 ± 1.9	12.6	8	304.9	−65.6	224.9	−66.2	162.8	4.4
MA16	60°07.09′	151°08.01′	277.1	18.0	18	8.4 ± 3.2	19.3	9	327.7	−58.6	295.5	−66.2	56.8	6.9
MA16								**8**	**316.8**	**−65.7**	**274.6**	**−68.2**	**268.2**	**3.4**
MA17	60°15.57′	151°04.25′	144.1	29.0	20	5.8 ± 8.3	35.2	8	67.5	66.2	230.0	86.7	10.3	18.1
MA18	60°15.62′	151°04.22′	121.1	19.0	12	15.4 ± 30.3	109.2	6	45.9	60.8	68.7	78.2	19.9	15.4
MA19	60°19.85′	151°11.72′	149.2	23.0	11	1.7 ± 0.37	2.4	8	50.8	69.2	289.8	86.1	392.2	2.8
Mean (in situ)								16	43.0	76.0			21.8	8.1
Mean (tilt corrected)							16			17.6	82.5	13.1	10.6	

The high-coercivity (normal text) and high-temperature (bold) magnetization components are listed in this table. Lat. and Long. correspond to latitude and longitude, respectively. *n* is the number of samples measured and used for calculation. Dec. and Inc. are declination and inclination, respectively; *k* is the Fisherian precision parameter (Fisher, 1953); α_{95} is the radius of cone at 95% confidence level about the mean direction. Sites with asterisks (MA03*, MA05* and MA13*) are excluded from mean direction calculation.

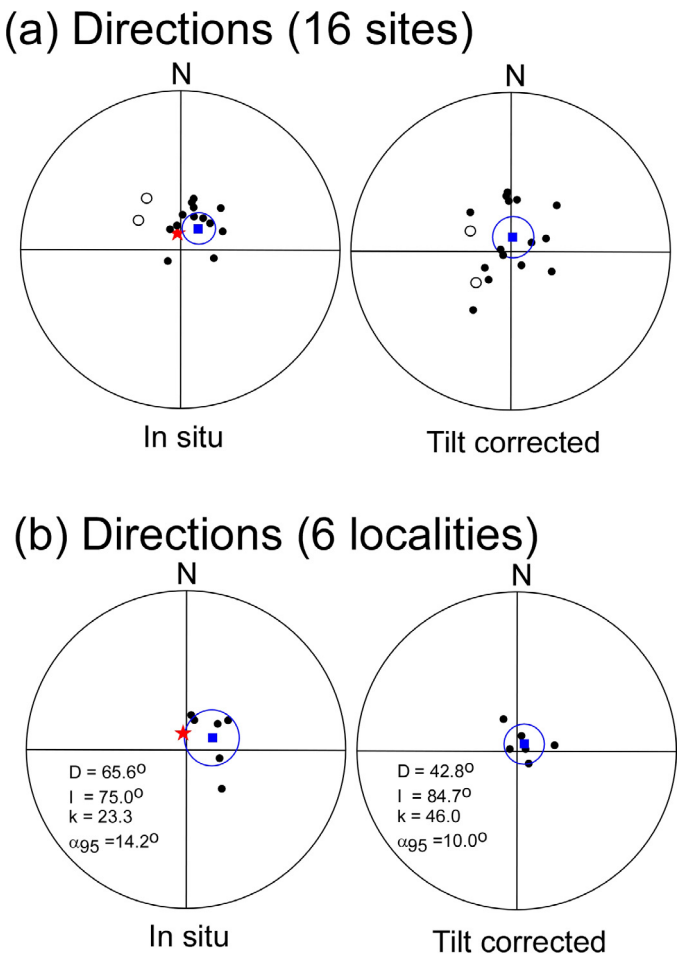


Fig. 7. Equal-area projections of (a) the site mean directions and (b) the locality mean directions. Formation mean directions are shown by blue squares with a 95% confidence circle. The axial dipole field direction is depicted by a red star. Solid (open) symbols are projections onto the lower (upper) hemisphere.

here for mean direction calculation. Positive fold tests and dual polarity behavior of these data sets thus suggest primary origin for the ChRM of the Kholchan and Ola suites.

As per the criteria of Van derVoo (1990), another genuine concern about the paleomagnetic results from basaltic flows is insufficient averaging of the paleosecular variation (PSV). However, we believe that formation mean direction calculated here on the basis of six localities, has sufficiently averaged out the PSV, because (1) the samples collected from the Kholchan and Ola suites represent a fairly long period of time between 85.5 Ma to 76.8 Ma, (2) the sampled localities cover an area of 30 km × 30 km, (3) the remanent directions reveal both normal and reversed polarities, and (4) the A_{95} value of 18.8° obtained from six locality-mean directions falls in a range between A_{95min} (8.3°) and A_{95max} (26.5°) which Deenen et al. (2011) declared reliable in their geomagnetic field model.

5.2. Poles comparison between the Kolyma-Omolon Composite Terrane and the Chukotka Terrane

A new Late Cretaceous (86–81 Ma) paleomagnetic pole calculated for the Magadan area by this study is considered as one of the representative poles for the Kolyma-Omolon Composite Terrane. Because of its geological juxtaposition with the Chukotka Terrane, the Late Cretaceous paleomagnetic poles from these terranes are compared here to discuss a tectonic connection between them (Fig. 8a).

On the basis of the locality-mean direction from Lake El'gygytgyn (67°20′ N, 172°00′ E), Stone et al. (2009) have determined a Late Cretaceous (90–67 Ma) paleomagnetic pole (69.2° N, 161.5° E, A_{95} = 16.5°) for the Chukotka Terrane (Fig. 8a). When compared with the same aged pole from the Kolyma-Omolon Composite Terrane (this study), an angular distance of just 3.6° (with 25.0° uncertainty at the 95% confidence level) appears between them. This similarity in pole positions is not unexpected because the respective data used for these poles come from the OCVB based igneous rocks which were emplaced on the Kolyma-Omolon and Chukotka terranes in the Early to Late Cretaceous. An agreement between these two poles, though at a low level of certainty, indicates no relative tectonic rotation/displacement between the Kolyma-Omolon and Chukotka terranes since 80 Ma (the time of OCVB formation). A tectonic model reported by Sokolov et al. (2002, 2009) estimates that a collision between the Kolyma-Omolon Composite Terrane and the Chukotka Terrane along the South Anyui Suture Zone had ceased by 115–119 Ma. We therefore claim that the composite Kolyma-Omolon-Chukotka Block behaved as a single tectonic unit since 80 Ma without any internal deformation. A characteristic 80 Ma paleomagnetic pole of 68.0° N, 164.7° E (A_{95} = 10.9°, N = 12) is thus calculated for the Kolyma-Omolon-Chukotka Block

Table 2
Paleomagnetic data for six localities from the Late Cretaceous Kholchan and Ola suites (Magadan area).

Sampling locality	Latitude	Longitude	n	In situ		k	α_{95} (°)	Tilt corrected		k	α_{95} (°)
				Dec. (°)	Inc. (°)			Dec. (°)	Inc. (°)		
Locality A MA01~MA08	60°19.75′	150°52.0′	6	15.3	74.3	36.7	11.2	16.4	81.9	7.6	26.1
Locality B MA09~MA12	60°18.20′	151°06.70′	4	9.3	72.1	93.0	9.6	336.4	72.3	17.5	22.6
Locality C MA14	60°19.26′	151°06.55′	1	105.1	72.4			136.6	81.0		
Locality D MA15 + MA16	60°07.08′	151°08.00′	2	137.5	62.6			80.2	70.2		
Locality E MA17 + MA18	60°15.60′	151°04.25′	2	55.7	63.9			75.7	85.6		
Locality F MA19	60°19.85′	151°11.72′	1	50.8	69.2			289.8	86.1		
Mean			6	65.6	75.0	14.2	23.3	42.8	84.7	46.0	10.0

Magnetization directions of the high-coercivity component are listed in this table. Columns headings are same as in Table 1.

Table 3
Paleomagnetic directions and pole positions from the Okhotsk-Chukotka Volcanic Belt (OCVB).

Terran	Study area		N	Palaeomagnetic pole		A_{95} (°)	dm (°)	dp (°)	
	Latitude	Longitude		Latitude (°N)	Longitude (°E)				
Kolyma-Omolon Superterrane									This study
Sites	60.4°N	151.0°E	16	72.4	165.7	18.9			
Localities	60.4°N	151.0°E	**6**	**66.7**	**168.5**	**18.8**			
Chukotka terrane									Stone et al. (2009)
Sites	67.3°N	172.0°E	40	67.0	171.0		9.8	9.8	
Localities	67.3°N	172.0°E	6	69.2	161.5		16.6	16.5	
Kolyma-Omolon-Chukotka Block									
Sites			56	68.3	169.0	8.1			
Localities			**12**	**68.0**	**164.7**	**10.9**			
Europe 80 Ma				81.0	232.5	6.9			Besse and Courtillot (2002)
Europe 80 Ma (calculated from 80 Ma paleoposition)				74.7	207.7	6.9			
East Asia 80 Ma				79.5	216.9	3.4			Cogné et al. (2013)
North America				74.7	207.4	5.9			Besse and Courtillot (2002)

Paleomagnetic poles are calculated using both the site-mean and locality-mean directions (localities). A_{95} is the radius of the cone at the 95% confidence level about the mean pole position. dp (dm) is a semi-axis of the confidence ellipse along the great–circle (site to pole). For comparison the 80 Ma pole positions from Europe and East Asia are also listed. Other explanations are the same as in Table 1.

after combining six data sets from the Magadan area (this study) and six from the Lake El'gygytgyn area (Fig. 8b).

5.3. Comparison between the Eurasian and Kolyma-Omolon-Chukotka poles

As shown in Fig. 8b, the 80 Ma pole for the composite Kolyma-Omolon-Chukotka Block is clearly distinguishable from the same age European and East Asian reference poles. The APWP of Besse and Courtillot (2002) places the 80 Ma European pole at the 81.4° N, 206.1° E (A_{95} = 5.9°) position. Using a compilation of paleomagnetic data from different blocks (including Siberia, Amuria, NCB, SCB and Korean Peninsula) Cogné et al. (2013) have calculated a new 80 Ma paleomagnetic pole (79.5° N, 216.9° E, A_{95} = 3.4°) for East Asia. When plotted together (Fig. 8b), the 80 Ma poles for Europe and East Asia fall on similar positions within the 95% confidence limit. However, the same age pole for the Kolyma-Omolon-Chukotka Block falls at 16.5–17.5° south of them.

This southerly pole position of the Kolyma-Omolon-Chukotka Block with respect to the same age poles from Europe and East Asia could either be due to changes in geomagnetic field direction or tectonic movement. Any deviation due to PSV of the geomagnetic field is discarded here because a reasonable smoothing of this phenomenon is ascertained by a wide distribution of paleomagnetic poles in the Kolyma-Omolon-Chukotka Block. In addition, we deduced an A_{95} value of 10.9° from 12 locality-based poles which falls in a reliable range between the A_{95min} value of 6.3° and the A_{95max} value of 17.1° (Deenen et al., 2011). A persistent non-dipolar component of the geomagnetic field can also cause steepness in inclination and a resultant divergence in pole positions. However, no such behavior has been observed in the Late Cretaceous rocks of Eastern Eurasia (Tsuneki et al., 2009; Cogné et al., 2013). Bearing ithis evidence in mind, we attribute the pole positions discrepancy between the Kolyma-Omolon-Chukotka Block and Europe/East Asia to tectonic activity rather than to geomagnetic field variations.

5.4. A possible post-80 Ma tectonic scenario of the Kolyma-Omolon-Chukotka Block

The above mentioned divergence in pole positions suggest that the composite Kolyma-Omolon-Chukotka Block acted independently from the main Eurasian Continent. We thus propose a new tectonic model in which the main Eurasian Continent is assumed to have experienced an eastward displacement as a result of opening in the Atlantic Ocean, while the Kolyma-Omolon-Chukotka Block underwent a southward displacement. As a consequence of these divergent displacements, the Kolyma-Omolon-Chukotka Block came in to collisional contact with Eurasia.

After assuming an eastward displacement for the main Eurasian Continent but no displacement for the Kolyma-Omolon-Chukotka Block since the Cretaceous, an 84 Ma pole of 74.7° N, 207.7° E is calculated for Europe (Fig. 8c). The same age paleopole for main Eurasia is then reconstructed by rotating it 19.7 °CCW about the Euler pole 66.5° N, 148.9° E (Srivastava et al., 1988; Muller and Roest, 1992; Gaina et al., 2002). Comparison of the 80 Ma pole from the Kolyma-Omolon-Chukotka Block with the 84 Ma pole of main Eurasia reveal an angular distance of 14.8 ± 12.4° between them. This comparison (Fig. 8c) also yields nearly identical positions for the reconstructed 84 Ma European pole and 80 Ma North American pole (i.e., 74.7° N, 207.4° E, A_{95} = 5.9°). However, the same age pole of the Kolyma-Omolon-Chukotka Block occupied a divergent position, suggesting a possible southward displacement (1640 ± 1380 km) of this block with respect to the North American and main Eurasian blocks.

This estimate of displacement indicates that 80 million years ago the Kolyma-Omolon-Chukotka Block was located at 75° N latitude as a part of the Arctic region. Nearlysimilar paleoposition has been predicted by some researchers (Lawver et al., 2002; Golonka et al., 2003; Golonka, 2007; Amato et al., 2004; Drachev and Sunders, 2006) for the Chukotka Terrane in the Early Cretaceous. We ascribe the post-80 Ma southward displacement of the Kolyma-Omolon-Chukotka Block to a more than 600 km opening of the Canadian Basin (Amerasian basin) in the Arctic region. Although several models have been proposed regarding the Canadian Basin opening using geological and geophysical observations (Lawver and Scotese, 1990; Lane, 1997; Grantz et al., 1998; Drachev and Sunders, 2006; Alvey et al., 2008), no clear regional magnetic anomaly pattern is reported yet for this basin to date the formation of the oceanic crust (Alvey et al., 2008; Saltus et al., 2011; Pease et al., 2014). The width of this basin, however, sheds some light on the 1640 ± 1380 km southward displacement of the Kolyma-Omolon-Chukotka Block (Fig. 8d).

The predicted southward displacement of the Kolyma-Omolon-Chukotka Block has probably caused its collision with the Siberian Craton (a part of the Eurasian Continent) and the ultimate formation of the peculiar "S" shaped Verkhoyansk and Chersky Ranges. A compressive regime induced by this collision has been documented by

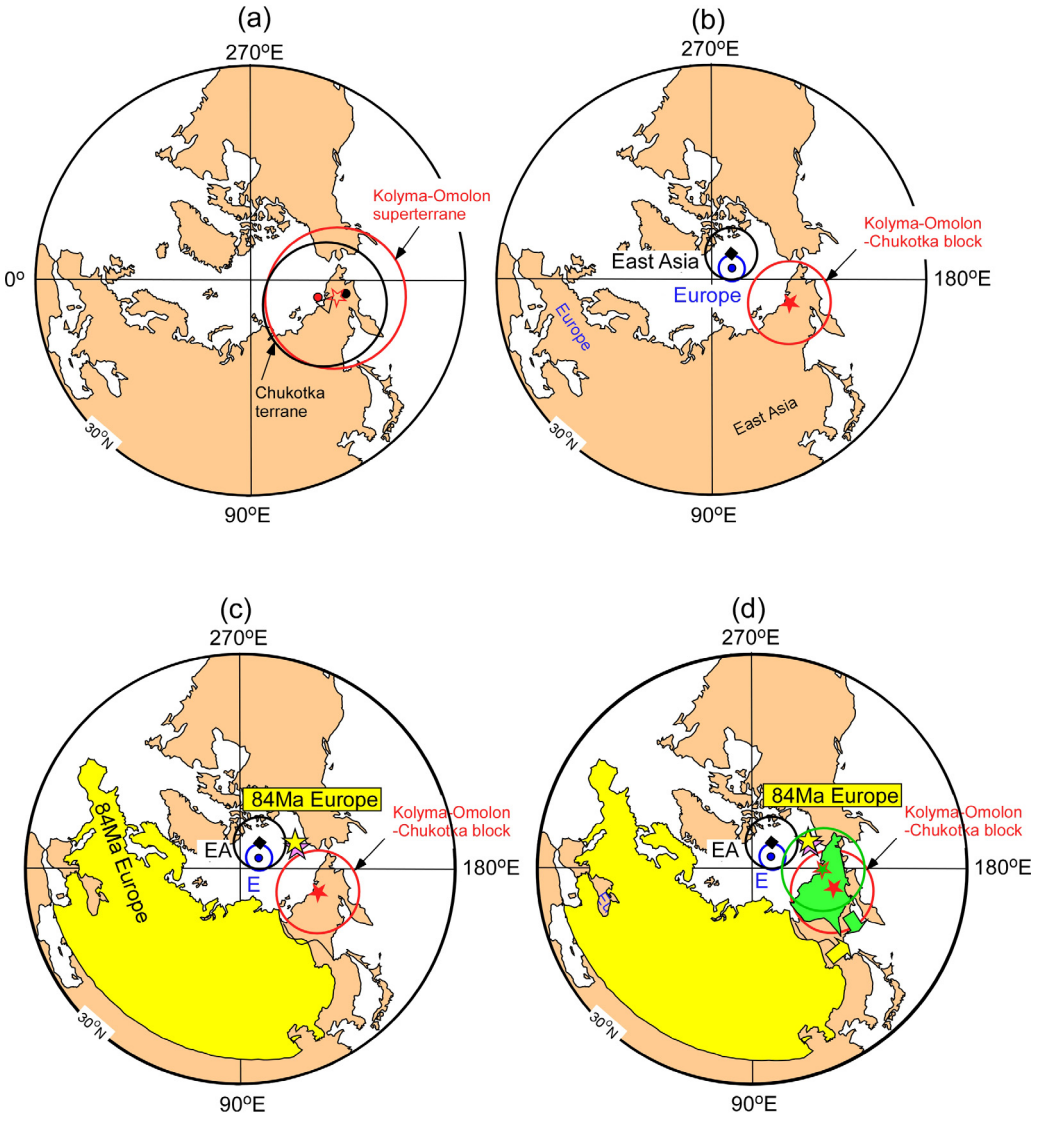


Fig. 8. (a) The paleomagnetic poles with 95% confidence circles for the Kolyma-Omolon Composite Terrane (open red star with red circle) and the Chukotka Terrane (triangle with black circle). These poles are calculated on the basis of locality-mean directions. The paleomagnetic poles calculated from site-mean directions are shown by small closed circles (red: Kolyma-Omolon Composite Terrane, black: Chukotka Terrane). (b) A paleomagnetic pole with 95% confidence circle for the Kolyma-Omolon-Chukotka Block (marked as a closed red star with red circle) is compared with the 80 Ma reference poles from Europe (Besse and Courtillot, 2002) and East Asia (Cogné et al., 2013). The European pole is marked by a dot with blue circle and the East Asian pole by a diamond with black circle. (c) The reconstructed 84 Ma paleoposition of the main Eurasian Continent is marked by a yellow star. The 80 Ma pole for North America is marked by a pink star. (d) The 80 Ma pole position of the Kolyma-Omolon-Chukotka Block (green star hemmed by red and a green 95% confidence circle) is located 600 km to north of the present position, assuming that southward displacement of this block is due to the opening of the Canadian Basin. After 80 Ma, the Kolyma-Omolon-Chukotka Block underwent a southward displacement (green arrow) and the main Eurasian Block moved eastward (yellow arrow). We predict a collision between the Kolyma-Omolon-Chukotka Block and Eurasia after 80 Ma as a consequence of these displacements. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

several researchers using deformational features in the Verkhoyansk fold-thrust belt (Parfenov, 1991; Sengor and Natalin, 1996; Layer et al., 2001; Oxman, 2003; Prokopiev and Oxman, 2009). In addition to this, on the basis of vitrinite reflectance analysis it has been suggested (Parfenov et al., 1995) that structural shortening in the Verkhoyansk fold-thrust belt started in the Early Jurassic and ended in the Late Cretaceous.

The post-80 Ma southward displacement of the Kolyma-Omolon-Chukotka Block predicted by this study, however, faces some contradiction with the reported geological interpretation. A geological constraint (Khanchuk et al., 2006) which goes against this displacement is the extent of OCVB in the Kolyma-Omolon-Chukotka Block and an overlay of its extreme western margin on the Siberian Craton. This problem is, however, overcome by the Early Cretaceous (142–120 Ma) paleomagnetic results reported from the Verkhoyansk depression of the Siberian Platform (Metelkin et al., 2008) which yields a pole position of 67.2° N, 183.8° E (A_{95} = 7.8°) and a resultant southward displacement of 500 km for this block compared to the same age poles from Europe. Comparison of this Siberian pole with the one determined for the Kolyma-Omolon-Chukotka Block (this study) yields a southward displacement of both of these terranes which solves the above noted problem regarding a geological contact between the OCVB and the Siberian platform. Nevertheless, we recommend the collection of additional geological, geo-chronological and paleomagnetic data from the OCVB to overcome any unclear issues regarding the post-80 Ma southward displacement of the Kolyma-Omolon-Chukotka Block.

6. Conclusions

Based on the new paleomagnetic results from the 86–81 Ma old ignimbrites of the Magadan area (the Kolyma-Omolon Composite

Terrane) we suggest the following conclusions about the northeast Eurasian tectonics.

(1) After successfully eliminating a lightning induced secondary magnetization by AFD procedure, the Late Cretaceous characteristic mean direction ($D = 42.8^\circ$, $I = 84.7^\circ$, $k = 46.0$, $\alpha_{95} = 10.0^\circ$) with a corresponding pole of 66.7° N, 168.5° E ($A_{95} = 18.8^\circ$) is estimated for the Magadan area.
(2) This new pole falls on the 67–90 Ma pole reported from Lake El'gygytgyn (Chukotka Terrane), revealing that the Kolyma-Omolon Composite Terrane and the Chukotka Terrane behaved as a single tectonic unit since 80 Ma. Accordingly a combined 80 Ma pole (68.0°, 164.7° E, $A_{95} = 10.9^\circ$) is calculated for the Kolyma-Omolon-Chukotka Block.
(3) This combined pole from 12 localities falls at 16.5–17.5° south of the coeval Eurasian poles, yielding a southward displacement of more than 260 km (1640 ± 1380 km) for the Kolyma-Omolon-Chukotka Block with respect to the main Eurasian and North American continents.
(4) Such large scale displacement of the Kolyma-Omolon-Chukotka Block could be ascribed to a post-80 Ma opening of the Canadian Basin in the Arctic Ocean.

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